

Research Design Concepts and Validity

Research Hypotheses

(and evidence required to support each)

Hypothesis Type	Evidence Required
1. Attributive	• A way to measure the behavior • How to discriminate it from related behaviors
2. Associative	• Reliable statistical relationship
3. Causal	• Temporal precedence (cause before effect) • Reliable statistical relationship • No confounds/alternative causal explanations

Research Design

Which participants do what when?

	Between Groups	Within Groups
True Experimental	BG True Exp.	WG True Exp.
Non-Experimental	BG Non Exp.	WG Non Exp.

Dimensions: Causal Interpretability (True Exp. vs. Non Exp.)

Participant Sampling Decisions

(and choices)

1. **Complete population** or **purposive** sampling frame
2. **Researcher selected** or **invited/self-selected**
3. **Simple** or **stratified**

Stages of Participant Sampling

1. **Target Population**
2. **Sampling Frame** (complete pop or purposive)
3. **Selected Sample** (researcher selected or self-selected & simple or stratified)
4. **Data Sample** (volunteerism & attrition)

Types of Participant Assignment

1. RA of individuals by the researcher
2. RA of intact groups
3. Arbitrary
4. Administrative
5. Self-assignment
6. Non-assignment ("measured IV")

Kinds of Measures/Behaviors

	Variable	Constant
Measured/Subject	Cell 1	Cell 2
Manipulated/Procedural	Cell 3	Cell 4

Role of each Measure/Behavior

1. **Causal/Independent variable**
2. **Effect/Dependent variable**

3. **Control**
 - variable
 - constant
4. **Confounding variable**

Types of Research Validity

1. **Measurement**
2. **External**
3. **Internal**
4. **Statistical Conclusion**

Internal Validity

(components and type of variables involved)

1. **Initial Equivalence** (measured/subject vars)
2. **Ongoing Equivalence** (manipulated/procedural vars)

External Validity

(components and type of variables involved)

1. **Population** (measured/subject)
2. **Setting** (manipulated/procedural vars)
3. **Task/Stimulus** (manipulated/procedural vars)
4. **Societal/Temporal**

Data Collection

1. **Collection** – observation, self-report or trace
2. **Setting** – laboratory, structured setting, or field
3. **Data source** – primary or archival

Research Validity Framework

Causal Relationship: IV causes DV

Three Requirements:
1. Temporal Precedence
2. Reliable statistical relationship
3. No alternative hypotheses/confounds

Measurement Validity

- Measurement error → bad data → worthless results

Key Questions:

- Is the IV properly manipulated?
- Is the DV properly measured?
- Are the values we have for every measure/behavior correct?

Data Types:

- Observational, Self-report or Trace?
- Primary or Archival data?

Every Characteristic/Behavior/Procedure

Type	Variable	Constant
Measured	1	2
Manipulated	3	4

Characteristic/Behavior Types & Their “Roles” in a Design

Role	Cell Type
Causal Variable (IV)	4
Effect Variable (DV)	2
Control Constant	1
Control Variable	2
Confounding Variable	2

Statistical Conclusion Validity

- IV & DV can't be causally related without statistical significance
- Statistical significance tests required

Programmatic Research

Choices we make influence Internal and External Validity:

Domain	Trade-off
Participants	representation vs. control
Setting	representation vs. control
Task-Stimulus	representation vs. control

Internal Validity

True Experimental vs. Non-experimental

Initial Equivalence – Participants:

- RA of individual participants before manipulation of the IV
- Without proper RA all subject variables are potential confounds
- Subject constants can't be confounds
- Subject variables that are controlled are control variables
- Subject variables that are not controlled are confounding variables (matching doesn't always work)

Ongoing Equivalence – Procedures:

- Only the IV is different across conditions
- Procedural constants can't be confounds
- Procedural variables that are the same across conditions are control variables
- Procedural variables that differ across conditions are confounding variables
- Ongoing equivalence is threatened by treatment implementation
- Ongoing equivalence is threatened by differences in procedures

External Validity

Population – Participant Sampling

Stage	Considerations
Target population	Who are you trying to generalize to?
Sampling Frame	Complete or Purposive
Selected Sample	Researcher-selected or Self-selected Simple or Stratified

Stage	Considerations
Data Sample	Attrition effects

Setting

- Laboratory, Structured or Field?

Task/Stimulus

- Familiar/Representative or Unfamiliar/Control?

Societal/Temporal

- Relationships among variables change over time in a society